

Convert the following algorithm into a program and find its time complexity using counter method.

```
void function(int n)
{
    int c= 0;
    for(int i=n/2; i<n; i++)
        for(int j=1; j<n; j = 2 * j)
            for(int k=1; k<n; k = k * 2)
                c++;
}
```

```
1 #include <stdio.h>
2
3 void function(int n)
4 {
5     int c = 0;
6
7     c++;
8
9     for (int i = n / 2; i < n; i++)
10    {
11        c++;
12
13        for (int j = 1; j < n; j = 2 * j)
14        {
15            c++;
16
17            for (int k = 1; k < n; k = k * 2)
18            {
19                c++;
20                c++;
21            }
22
23            c++;
24        }
25
26        c++;
27    }
28
29    c++;
30
31    printf("%d", c);
32 }
33
34 int main()
35 {
36     int n;
37     scanf("%d", &n);
38
39     function(n);
40
41     return 0;
42 }
```

	Input	Expected	Got	
✓	4	30	30	✓
✓	10	212	212	✓

Passed all tests! ✓

Correct

Marks for this submission: 1.00/1.00.